

Joseph Paul Koyi

Profile

Computer science graduate with a strong foundation in software engineering, specializing in high-performance systems. Passionate about solving complex problems programmatically and building production-grade software.

Skills

Languages Java, C++, Python, C, SQL, NOSQL

Core Concepts Algorithms, Data Structures, Complexity Analysis, Concurrency, OOP

Systems Performance Optimization, Memory Management, Low-Latency Design

Tools Git, Docker, JUnit, CMake, GitHub Actions, Linux

Practices Full SDLC, Code Review, CI/CD, TDD

Work Experience

Nov 2024–May 2025 **Software Engineering Intern**, *Ericsson*, Budapest, Hungary

- Built static analyzers for Java using SpotBugs and SEI CERT secure coding standards, achieving 99% true positive rate
- Designed and tested custom bug detectors with high precision, validated on large-scale open-source Java projects
- Implemented detector registration, XML configurations, and contributed to GitHub/GitLab PR workflows

Projects

NebulaPower Platform

- Built a high-throughput data processing platform in Java Spring Boot for real-time monitoring and analytics. Implemented concurrent stream processing for multiple data sources, scheduled task execution for automated compliance checks, and asynchronous event-driven alerting. Designed for performance with JPA query optimization and PostgreSQL indexing strategies.

BareMetalVM

- Engineered a low-level 16-bit virtual machine in C with custom instruction set architecture and register-based execution. Implemented memory-efficient variable-length instruction encoding, direct memory access with bounds checking, and stack-based execution model. Demonstrates deep understanding of systems programming, memory management, and performance-critical code.

Education

Sept 2023–Jan 2026 **BSc Computer Science**, *Eötvös Loránd University*, Budapest, Hungary

Relevant coursework: Algorithms & Data Structures, OOP, Operating Systems, Concurrent Programming, Computer Architecture

Certifications

- Docker Foundations Professional Certificate – Docker Inc. (Feb 2025)
- Ubuntu Linux Professional Certificate – Canonical (Feb 2025)
- Java Foundations Professional Certificate – JetBrains (Feb 2025)